

Regulation of MHC I Molecules in Glioblastoma Cells and the Sensitizing of NK Cells.

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Immunotherapy has been established as an important area in the therapy of malignant diseases. Immunogenicity sufficient for immune recognition and subsequent elimination can be bypassed by tumors through altered and/or reduced expression levels of major histocompatibility complex class I (MHC I) molecules. Natural killer (NK) cells can eliminate tumor cells in a MHC I antigen presentation-independent manner by an array of activating and inhibitory receptors, which are promising candidates for immunotherapy. Here we summarize the latest findings in recognizing and regulating MHC I molecules that affect NK cell surveillance of glioblastoma cells.

Strategies to Augment Natural Killer (NK) Cell Activity against Solid Tumors.

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The immune system plays a crucial role to prevent local growth and dissemination of cancer. Therapies based on activating the immune system can result in beneficial responses in patients with metastatic disease. Treatment with antibodies targeting the immunological checkpoint axis PD-1 / PD-L1 can result in the induction of anti-tumor T cell activation leading to meaningful long-lasting clinical responses. Still, many patients acquire resistance or develop dose-limiting toxicities to these therapies. Analysis of tumors from patients who progress on anti-PD-1 treatment reveal defective interferon-signaling and antigen presentation, resulting in immune escape from T cell-mediated attack. Natural killer (NK) cells are innate lymphocytes that can kill tumor cells without prior sensitization to antigens and can be activated to kill tumor cells that have an impaired antigen processing and presentation machinery. Thus, NK cells may serve as useful effectors against tumor cells that have become resistant to classical immune checkpoint therapy. Various approaches to activate NK cells are being increasingly explored in clinical trials against cancer. While clinical benefit has been demonstrated in patients with acute myeloid leukemia receiving haploidentical NK cells, responses in patients with solid tumors are so far less encouraging. Several hurdles need to be overcome to provide meaningful clinical responses in patients with solid tumors. Here we review the recent developments to augment NK cell responses against solid tumors with regards to cytokine therapy, adoptive infusion of NK cells, NK cell engagers, and NK cell immune checkpoints.

Adoptive transfer of autologous natural killer cells leads to high levels of circulating natural killer cells but does not mediate tumor regression.

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Adoptive transfer of tumor-infiltrating lymphocytes (TIL) can mediate regression of metastatic melanoma. However, many patients with cancer are ineligible for such treatment because their TIL do not expand sufficiently or because their tumors have lost expression of antigens and/or MHC molecules. Natural killer (NK) cells are large granular lymphocytes that lyse tumor cells in a non-MHC-restricted manner. Therefore, we initiated in a clinical trial to evaluate the efficacy of adoptively transferred autologous NK cells to treat patients with cancers who were ineligible for treatment with TIL. Patients with metastatic melanoma or renal cell carcinoma were treated with adoptively transferred in vitro activated autologous

NK cells after the patients received a lymphodepleting but nonmyeloablative chemotherapy regimen. Clinical responses and persistence of the adoptively transferred cells were evaluated. Eight patients were treated with an average of 4.7×10^{10} ($\pm 2.1 \times 10^{10}$) NK cells. The infused cells exhibited high levels of lytic activity in vitro. Although no clinical responses were observed, the adoptively transferred NK cells seemed to persist in the peripheral circulation of patients for at least one week posttransfer and, in some patients, for several months. However, the persistent NK cells in the circulation expressed significantly lower levels of the key activating receptor NKG2D and could not lyse tumor cell targets in vitro unless reactivated with IL-2. The persistent NK cells could mediate antibody-dependent cell-mediated cytotoxicity without cytokine reactivation in vitro, which suggests that coupling adoptive NK cell transfer with monoclonal antibody administration deserves evaluation.

On-Chip Biogenesis of Circulating NK Cell-Derived Exosomes in Non-Small Cell Lung Cancer Exhibits Antitumoral Activity.

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As the recognition between natural killer (NK) cells and cancer cells does not require antigen presentation, NK cells are being actively studied for use in adoptive cell therapies in the rapidly evolving armamentarium of cancer immunotherapy. In addition to utilizing NK cells, recent studies have shown that exosomes derived from NK cells also exhibit antitumor properties. Furthermore, these NK cell-derived exosomes exhibit higher stability, greater modification potentials and less immunogenicity compared to NK cells. Therefore, technologies that allow highly sensitive and specific isolation of NK cells and NK cell-derived exosomes can enable personalized NK-mediated cancer therapeutics in the future. Here, a novel microfluidic system to collect patient-specific NK cells and on-chip biogenesis of NK-exosomes is proposed. In a small cohort of non-small cell lung cancer (NSCLC) patients, both NK cells and circulating tumor cells (CTCs) were isolated, and it is found NSCLC patients have high numbers of NK and NK-exosomes compared with healthy donors, and these concentrations show a trend of positive and negative correlations with bloodborne CTC numbers, respectively. It is further demonstrated that the NK-exosomes harvested from NK-graphene oxide chip exhibit cytotoxic effect on CTCs. This versatile system is expected to be used for patient-specific NK-based immunotherapies along with CTCs for potential prognostic/diagnostic applications.

Natural killer cells: a review of biology, therapeutic potential and challenges in treatment of solid tumors.

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Natural killer (NK) cells lead immune surveillance against cancer and early elimination of small tumors. Owing to their ability to engage tumor targets without the need of specific antigen, the therapeutic potential of NK cells has been extensively explored in hematological malignancies. In solid tumors, however, their role in the clinical arena remains poorly exploited despite a broad accumulation of preclinical data. In this article, we review our current knowledge of NK cells' biology, and highlight the challenges facing NK cell antitumor strategies in solid tumors. We further summarize the abundant preclinical attempts at overcoming these challenges, present past and ongoing clinical trial data and finally discuss the potential impact of novel insights on the development of NK cell-based therapies.

NK cell therapy for hematologic malignancies.

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Natural killer (NK) cells are part of the innate immune system and represent the first line of defense against infections and tumors. In contrast to T cells, NK cells do not require prior antigen sensitization to induce cytotoxicity and do not cause graft-versus-host disease. These, along with other advantages, make NK cells an attractive candidate for adoptive cellular therapy. Herein, we describe the mechanisms of NK cell cytotoxicity, which is governed by an intricate balance between various activating and inhibitory receptors, including the killer cell immunoglobulin-like receptors (KIRs). We illustrate the advantages of NK alloreactivity as demonstrated in various types of hematopoietic stem cell transplants (HSCT), such as haploidentical, human leukocyte antigen-matched related or unrelated donor and umbilical cord blood transplant. We elaborate on different models used to predict NK cell alloreactivity in these studies, which are either based on the absence of the ligands for inhibitory KIRs, presence of activating NK cell receptors and KIR genes content in donors, or a combination of these. We will review clinical studies demonstrating anti-tumor efficacy of NK cells used either as a stand-alone immunotherapy or as an adjunct to HSCT and novel genetic engineering strategies to improve the anti-tumor activity of NK cells.

[Clinical relevance of KIRs in hematopoietic stem cell transplantation].

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Natural Killer cells (NK cells) represent the subset of peripheral lymphocytes that play critical role in the innate immune response to virus-infected and tumor transformed cells. Lysis of NK sensitized target cells could be mediated independently of antigen stimulation, and unlike cytotoxic T-lymphocytes, they do not require peptide presentation by the major histocompatibility complex (MHC) molecules. NK cell cytotoxic activity is controlled by considerable number of cell surface Killer cell Immunoglobulin like Receptors (KIRs), which can exist in both inhibitory and activating isoforms. The inhibitory KIRs are mostly specific for HLA class I ligands and I HLA class like molecules, while the specificity of activating receptors is regarded to lectine-like superfamily. The role of NK cells in allogeneic haematopoietic stem cell transplantation (HSCT): NK cells are the first lymphocyte subset that reconstitute the peripheral blood following allogeneic HSCT. By selecting donors mismatched for relevant HLA ligands in the context of recipients KIR genotype, multiple roles for alloreactive donor NK cells have been demonstrated, in diminishing Graft vs. Host Disease (GvHD) through selective killing of recipient dendritic cells, prevention of graft rejection by killing recipient T cells and participation in Graft vs. Leukaemia (GvL) effect through destruction of residual host tumor cells. Investigation of KIRs heterogeneity play an important role in the field of HSCT, because it is useful for the early diagnosis of post transplant complications and can serve as a predictive risk factor for GvHD development.

Mechanisms of tumor and viral immune escape from natural killer cell-mediated surveillance.

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Human natural killer (NK) cells recognize and efficiently eliminate MHC class I low or negative malignant targets and virally infected host cells, without requirement for prior sensitization. However, viruses and various tumor cells display elaborate adaptations to evade and overcome immunosurveillance. The current review focuses on escape mechanisms of viruses and malignantly transformed 'stressed' cells to evade from NK cell cytotoxicity. A general overview of recent clinical studies using allogeneic donor NK cells is given, summarizing first data about a possible benefit for patients suffering from high-risk leukemia and solid tumors. Finally, the review discusses the future perspectives and hypotheses aiming to improve therapeutic NK cell strategies against tumor immune escape mechanisms.

Cross-Talk Between Tumor Cells Undergoing Epithelial to Mesenchymal Transition and Natural Killer Cells in Tumor Microenvironment in Colorectal Cancer.

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Tumor cells undergoing epithelial to mesenchymal transition (EMT) and immune cells in tumor microenvironment (TME) reciprocally influence each other. Immune cells, by supplying TME with bioactive molecules including cytokines, chemokines, enzymes, metabolites, and by physical interactions with tumor cells via their receptors, represent an important factor that affects EMT. Chronical inflammation in TME favors tumor growth and invasiveness and stimulates synthesis of EMT promoting transcription factors. Natural killer (NK) cells, owing to their unique ability to exert cytotoxic function independent of major histocompatibility (MHC)-mediated antigen presentation, play a significant role in the control of metastasis in colorectal cancer (CRC). Although, the cross-talk between immune cells and tumor cells in general favors the induction of EMT and inhibition of antitumor immune responses, there are some changes in the immunogenicity of tumor cells during EMT of CRC cells that increase their susceptibility to NK cell cytotoxic lysis. However, suppressive TME downmodulates the expression of activating NK cell receptors, decreases the expression of activating and increases the expression of inhibitory NK cell ligands on tumor cells, and impairs NK cell metabolism that altogether negatively affects the overall NK cell function. Furthermore, process of EMT is often associated with increased expression of programmed cell death ligand (PD-L) and expression of immune checkpoint molecules PD-1, TIGIT, and TIM3 on functionally exhausted NK cells in TME in CRC. In this review we discuss modalities of cross-talk between tumor cells and NK cells, with regard of EMT-driven changes.

Interferon Gamma Induces Changes in Natural Killer (NK) Cell Ligand Expression and Alters NK Cell-Mediated Lysis of Pediatric Cancer Cell Lines.

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Natural killer (NK) cells have therapeutic potential for cancer due to their capacity for targeting tumor cells without prior sensitization. Our laboratory has developed an NK cell expansion protocol that generates large quantities of NK cells for therapeutic infusion that secrete 20 times the amount of interferon gamma (IFN γ) than resting NK cells. IFN γ can upregulate major histocompatibility complex (MHC)-class I, an inhibitory ligand for NK cells, but can also upregulate intercellular adhesion molecule 1 (ICAM-1) which promotes NK:target cell interaction for an efficient lysis. Due to the opposing effects reported for IFN γ on tumor sensitivity to NK cells, we evaluated a panel of 22 tumor cell lines from the pediatric preclinical testing program corresponding to different tumor types. We determined the impact

of IFN γ on their expression of NK cell activating and inhibitory ligands, death receptors, and adhesion molecules using mass cytometry. We also evaluated the effect of IFN γ on their sensitivity to NK cell-mediated lysis. Our results show upregulation of PD-L1, ICAM-1, MHC-class I, HLA-DR, CD95/FasR, and CD270/HVEM after IFN γ treatment, this upregulation is variable across different tumor types. We also observed a variable impact of IFN γ in NK cell-mediated lysis. For six of the cancer cell lines IFN γ resulted in increased resistance to NK cells, while for three of them it resulted in increased sensitivity. Modeling of the data suggests that the effect of IFN γ on NK cell-mediated tumor lysis is mostly dependent on changes in MHC-class I and ICAM-1 expression. For three of the cell lines with increased resistance, we observed higher upregulation of MHC-class I than ICAM-1. For the cell lines with increased sensitivity after IFN γ treatment, we observed upregulation of ICAM-1 exceeding MHC-class I upregulation. ICAM-1 upregulation resulted in increased conjugate formation between the NK cells and tumor cells, which can contribute to the increased sensitivity observed. However, the effects of MHC-class I and ICAM-1 are not readily predictable. Due to the high IFN γ secretion of NK cell infusion products, a better understanding of the NK ligands on tumor cells and how they are affected by IFN γ is essential to optimize NK cell immunotherapy.
